PACKET — PAUSA MX Week 12

Founder: Miguel Bravo
Course: AI 2041 — Week 12 — Extraction + the core build
Selected candidate: PAUSA MX (explored in Week 10)
Language of product UI: Mexican Spanish
Documentation language: English, except quoted product copy

Provisional methodology. METHOD, VERDICT, MANIFESTO, and the Week 12 team BLUEPRINT were not in the workspace. See `docs/SOURCE_REVIEW.md`.

Step 3 of my Method becomes the function that converts a proposed purchase, existing obligations and available money into a transparent comparison of the user's financial position before and after buying.

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1. Problem in the founder's words

Many people in Mexico judge a purchase by the number on the product screen. That number is not the same as what the purchase does to money that is already claimed by rent, food, transport, and payments already promised. Installment tables make the monthly quote look small and leave the total actually committed, and the cash left today, off-stage. PAUSA MX exists so someone can pause and see those constraints before they buy. It does not approve credit, connect a bank, or tell them they are good or bad with money.

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2. Exact user

A young adult in Mexico who is considering a consumer purchase, can type a few amounts in MXN, and wants to understand the effect on available money. Primary design hypotheses: Valeria (23, CDMX, salary, phone, full payment) and Daniel (27, variable income, computer, installments). Both are synthetic. See `docs/PERSONAS.md`.

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3. Definition of success before the module closes

The module is successful for Week 12 if **all** of the following are true:

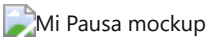
1. Charter, personas, and packet exist in git **before** this repository's application rebuild commits.
2. One working feature, Mi Pausa, computes a consistent comparison from typed inputs.
3. Unit tests cover the required mechanical cases.
4. Private save requires Google authentication via Supabase; RLS restricts rows to the owner.
5. Public Vercel deployment exists. A real purchased domain is connected **or** the founder has exact purchase instructions (this agent cannot buy a domain).
6. Documentation states what was tested and what was not invented.

Success is **not** a claimed 10x impact, a paying customer, or a completed external persona-test chat until those actually happen.

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4. Visual mockup

Generated with Cursor image generation on 2026-11-01 and saved as `docs/mockup.png`. It is a product UI illustration of Mi Pausa (form + Comprar ahora / Esperar). It is not a screenshot of the production app.


Mi Pausa mockup

5. Mermaid flowchart

```
flowchart TD
  A[Home: Antes de comprar, haz una pausa] --> B[Mi Pausa form]
  B --> C{Inputs valid?}
  C -->|No| D[Show field errors. No invented numbers]
  C -->|Yes| E[Deterministic engine in TypeScript]
  E --> F[Money after obligations]
  E --> G[Immediate cost]
  E --> H[Future payments created]
  E --> I[Remaining after immediate purchase]
  E --> J[Explanation + future-affordability warning if needed]
  E --> K[Comparison: Comprar ahora vs Esperar]
  K --> L{User wants to save?}
  L -->|No| M[Keep preview local. Do not persist amounts]
  L -->|Yes| N{Authenticated with Google?}
  N -->|No| O[Prompt sign-in. Do not store yet]
  N -->|Yes| P[Insert evaluation with user_id. RLS]
  P --> Q[Reopen from saved list]
```

6. Mermaid sequence diagram

```
sequenceDiagram
  actor Person
  participant UI as Mi Pausa UI
  participant Engine as evaluatePurchase()
  participant Auth as Supabase Auth
  participant DB as PostgreSQL + RLS

  Person->>UI: Enters MXN amounts and payment plan
  UI->>UI: Zod validation client-side
  UI->>Engine: Typed numbers only
  Engine-->>UI: Comparison + explanation + warnings
  Note over UI,Engine: Preview stays in the browser. No POST of amounts.
  Person->>UI: Guardar evaluación
  UI->>Auth: Google session?
  alt Not signed in
    Auth-->>UI: Challenge
    UI-->>Person: Sign in to save. Preview remains local.
  else Signed in
    UI->>DB: Insert evaluation (user_id from JWT)
    DB-->>UI: Row if user_id = auth.uid()
    Person->>UI: Open saved evaluation
    UI->>DB: Select own rows
    DB-->>UI: Owner rows only
  end
```

7. Global benchmark and differentiation

Comparable product (verified pages): BBVA México publishes consumer financial-education calculators, including "Interactivos dinero inteligente":

- https://www.bbva.mx/educacion-financiera/interactivos-dinero-inteligente.html

That page (fetched 2026-11-01) presents simulators such as a 70/20/10 administration method and a calculator "para practicar cómo organizar mejor tu dinero, controlar tus gastos y planear tu ahorro."

A related BBVA budget calculator (BBVA corporate site, Spanish, fetched 2026-11-01) is:

- https://www.bbva.com/es/salud-financiera/calculadora-presupuestos/

Verified from that page: it asks for monthly net income and categories of expenses (obligatory fixed, reducible fixed, and easier-to-change spending). It states: "Tus datos están seguros. Este servicio no almacenará la información que introduzcas."

Assumptions (not verified on this pass): that the BBVA México interactive calculators persist nothing server-side; that they do not compare a specific contemplated purchase against existing commitments; that they are education tools attached to a bank brand.

Differentiation (product intent, not a competitive study): PAUSA MX is purchase-decision shaped (one contemplated buy, obligations, payment plan as typed, Comprar ahora vs Esperar). It is not a bank product, does not recommend BBVA or any lender, and does not invent a 50/30/20 or 70/20/10 split.

Week 10 PAUSA MX is an ancestor prototype (suggestion engine, localStorage). This rebuild removes the buy/wait command and adds authenticated persistence.

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8. Three-sentence long view

If people repeatedly use a pause that makes constraints visible, they may buy fewer things they cannot carry — or buy the same things with fewer surprises. The venture can later charge for saved history, shared household context, or a documented export, only after someone other than the founder says those are worth paying for. Until then the 10:1 hypothesis is a claim to test, not a result.

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9. Scope cuts

Out of Week 12:

- Merchant catalog, checkout, BNPL partnerships
- Bank aggregation
- Multi-obligation calendar (Week 10 `/pagos`)
- Verbal recommendation engine (Week 10 outcomes)
- Chatbot
- Custom domain purchase performed by the agent
- Real user interviews
- Analytics suite for the north star

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10. Architecture and stack

Layer	Choice	Why

App	Next.js App Router	Required; server actions for save
UI	React + Tailwind CSS	Required; mobile-first
Language	TypeScript strict	Testable engine
Validation	Zod	Client and server
Engine	Pure functions in `src/engine`	Independent of UI
Auth	Supabase Auth, Google OAuth	Required for private save
Data	Supabase PostgreSQL + RLS	Owner-only evaluations
Tests	Vitest + Playwright	Engine + smoke
Host	Vercel	Required
Repo	GitHub	Required
Secrets	`.env.local` uncommitted; `.env.example` committed	No service-role in browser

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11. Mechanical test plan

See also `docs/TEST_PLAN.md` (created with the test suite). Required cases:

1. Valid full-payment purchase
2. Purchase exceeding available money
3. Zero available money
4. Installment plan with valid values
5. Invalid installment data
6. Negative amounts
7. Extremely large values
8. Decimal rounding
9. Empty required fields
10. Unauthenticated save attempt
11. Authenticated save and reopen
12. RLS isolation between two test users

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12. Persona test plan

A synthetic persona test of Valeria or Daniel must run in a **fresh LLM chat** with real screenshots of the working app. This packet does not contain that test. Script and evidence log: `docs/PERSONA_TEST.md`.

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13. Security plan

- Google OAuth required to persist evaluations
- RLS: `user_id = auth.uid()` for select/insert/update/delete
- Anon key in browser; **service-role key never in client**
- Zod on client and server; numeric range 0–10,000,000.00 MXN; installment count 2–60
- Preview mode: compute in-browser only
- Privacy page in Mexican Spanish explaining data use

- Secret scan before each push

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14. Genuine commit and deployment plan

1. Charter, personas, packet (this commit family)
2. Project architecture and interface
3. Deterministic engine and unit tests
4. Supabase auth, persistence, RLS
5. QA, docs, deployment readiness

Push to GitHub when authenticated. Deploy to Vercel at least twice. Domain purchase is a founder step.

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15. Evidence requirements

Allowed as evidence only if real: git log, test runner output, Vercel deployment IDs, public HTTP checks, SQL migration files, screenshots of this app, Cursor-generated mockup file.

Not allowed to fabricate: interviews, team votes, domain purchase, DEMO video, BUILDCHAT export, persona-test results, 10x impact, invented bugs.

Engine sketch (normative)

Let amounts be decimal MXN converted to integer cents with half-up rounding on the third decimal of pesos (not JavaScript `Math.round(mxn * 100)`, which fails for values such as `1.005`).

- `moneyAfterObligations = availableMoney − essentialExpenses − committedPayments`
- `immediateCost = full payment ? purchasePrice : downPayment`
- `futurePaymentsTotal = full payment ? 0 : installmentCount × installmentAmount`
- `remainingAfterImmediate = moneyAfterObligations − immediateCost`
- `waitRemaining = moneyAfterObligations`
- `waitFutureFromThisPurchase = 0`
- If installments: `enteredPurchaseTotal = downPayment + futurePaymentsTotal`; `differenceVsCashPrice = enteredPurchaseTotal − purchasePrice`
- Future affordability warning when payment method is installments **or** when any future amount is created, because the form does not collect future income.

The Comparison presents these figures. It does not select a winner.